Determination of the Neutrino Time of Flight from the NuMI Target to the NOvA Detectors: Coordinate Systems, Distances and Timing Offsets

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Abstract

This note derives, in full, the straight-line distances between the NuMI target, the NOvA Near Detector (ND), the NOvA Far Detector (FD) at Ash River and the MINOS Far Detector at Soudan, and the corresponding neutrino times of flight in SI units and in 64 MHz NOvA clock ticks. Five coordinate systems are used and written out explicitly: a spherical Earth, WGS84 geodetic coordinates and the Earth-centred Earth-fixed (ECEF) frame, the Fermilab Site Coordinate System (a double stereographic projection), the NuMI beamline system with origin MCZERO in which the Fermilab alignment group surveyed both NOvA upstream-face centres, and a beam-axis frame used for off-axis and opening angles. The target is taken from the 1998 NuMI Technical Design Report beam sheet; the detector endpoints are the surveyed face centres. Every correction between the reference model and the final result is tabulated, together with chord geometry, the Main Injector timing shift, the trigger window, relativistic and instrumental terms, sensitivities and an error budget. The headline result is a flight time from the ND face to the FD face of 2,700.250552 µs, i.e. 172,816.035 ticks, and a total offset including the 141-turn Main Injector shift of 272,736 ticks. The 2013 Mathematica notebook and the 2014 note [1] are reproduced to 10^{-9} and their superseded geometries are compared with the surveyed result in an appendix. All computations are performed by the `gps-tof` package [2]; the tables of this note are generated from its JSON report.

Contents

1 Introduction **2**

2 Reference frames and transformations **3**

2.1 Angles 3
2.2 Geodetic coordinates and the ECEF frame 3
2.3 Spherical-Earth model 4
2.4 Local frame and antenna-to-detector offsets 4
2.5 Fermilab Site Coordinate System 4
2.6 NuMI beamline system (origin MCZERO) and the beam-axis frame 5

3 Input positions **6**
3.1 NuMI target, MCZERO and the design beam axis 6
3.2 Surveyed upstream-face centres of the NOvA detectors 6
3.3 GPS antenna fixes, MINOS far detector and other inputs 7
3.4 Converted positions 7

4 Distances **7**
4.1 Chords in the spherical and ellipsoidal models 7
4.2 Correction ledger 7
4.3 Chord geometry 9

5 Time of flight and timing offsets **11**
5.1 Flight time and clock ticks 11
5.2 Main Injector timing shift 11
5.3 Combined offset and trigger window 11
5.4 Effects outside the geometric flight time 12

6 Angles relative to the beamline **12**
6.1 Beam axis and off-axis positions 12
6.2 Opening angles 12
6.3 Near Detector face edges 14

7 Validation **14**

8 Sensitivities and error budget **14**

9 Summary **17**

A History and reproduction of the earlier calculations **17**

B Software and reproducibility **17**

1 Introduction

The NOvA data acquisition issues a "beam" trigger from the NuMI spill time t_0 delayed by the neutrino flight time t_{delay} from the beam source to the far detector. The spill is nominally 10 µs long; the DAQ discretises data into 50 µs microslices and the event builder extracts whole microslices, so the trigger window is the half-open interval [1]

$$\Delta T_{\mathrm{trig}} = [\, t_0 + t_{\mathrm{delay}} - 20\,\mu\mathrm{s},\ t_0 + t_{\mathrm{delay}} + 30\,\mu\mathrm{s}\,)\,, \tag{1}$$

which always overlaps exactly two closed microslices. t_{delay} has to be set a priori from the geometry.

The 2013 Mathematica notebook [3] computed the chord between the MINOS surface-building GPS antenna, lowered by 300 m, and an Ash River antenna. The 2014 note NOvA-doc-11860 [1]

documented the antenna fixes and specified antenna-to-detector offsets. Neither contained the NuMI target, and the horizontal antenna-to-detector vectors were left to be determined. This note (i) derives the target position from the NuMI Technical Design Report [4] through the Fermilab site projection [5], (ii) records the surveyed upstream-face centres of both NOvA detectors [6, 7] in the NuMI beamline system and converts them to geodetic and ECEF coordinates, (iii) writes out every coordinate transformation used, and (iv) tabulates the distances, flight times, angles and uncertainties. Figures 1–4 are schematic views of the geometry.

2 Reference frames and transformations

2.1 Angles

Survey records give angles in sexagesimal form. Degrees, minutes and seconds are converted to radians by

$$\mathrm{LC}(d, m, s) = \frac{\pi}{180}\left(d + \frac{m + s/60}{60}\right), \tag{2}$$

with the hemisphere letter supplying the sign. Longitudes are east-positive throughout; the 2013 notebook entered west longitudes as positive numbers, which leaves distances unchanged (a reflection is an isometry) but flips the sign of rotation-dependent terms such as the Sagnac correction (Sec. 5.4).

2.2 Geodetic coordinates and the ECEF frame

The reference ellipsoid is WGS84 [8], with semi-major axis a, semi-minor axis $b = a(1 - f)$ and first eccentricity $e^2 = (a^2 - b^2)/a^2$:

$$\begin{aligned} a &= 6{,}378{,}137\,\mathrm{m}, & 1/f &= 298.257223563, \\ b &= 6{,}356{,}752.31424518\,\mathrm{m}, & e^2 &= 6.69437999 \times 10^{-3}. \end{aligned} \tag{3}$$

The Fermilab site survey is in NAD83 on the GRS80 ellipsoid ($1/f = 298.257222101$), which differs from WGS84 by 0.1 mm in b; the two are treated as coincident, as the Fermilab coordinate-system document itself does [5]. A point with geodetic latitude φ, longitude λ and ellipsoidal height h maps to Earth-centred Earth-fixed Cartesian coordinates by

$$N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}}, \qquad \begin{aligned} X &= (N + h)\cos\varphi\cos\lambda, \\ Y &= (N + h)\cos\varphi\sin\lambda, \\ Z &= \big(N(1 - e^2) + h\big)\sin\varphi. \end{aligned} \tag{4}$$

N is the prime-vertical radius of curvature. The X axis passes through the intersection of the equator and the prime meridian, Z through the pole. The inverse transformation is evaluated iteratively to 10^{-14} rad. The height h in Eq. (4) is the height above the ellipsoid. A GPS receiver's `$GPGGA` altitude is orthometric (above mean sea level), H; the same sentence gives the geoid separation N_g and $h = H + N_g$. Each input position in this note carries an explicit datum, and the conversion appears as a line in the correction ledger.

2.3 Spherical-Earth model

As a reference model, used only to show the size of the ellipsoidal correction, the Earth is taken as a sphere of radius $R = a$ and

$$\vec{r} = (R + h)\,(\cos\varphi\cos\lambda,\ \cos\varphi\sin\lambda,\ \sin\varphi). \tag{5}$$

A sphere of equatorial radius places both sites about 10 km too far from the Earth's centre, and since the chord scales with the radius the Fermilab–Ash River chord comes out 823 m (176 ticks) too long (Sec. 4.1).

2.4 Local frame and antenna-to-detector offsets

At a point (φ, λ) the local East, North and Up unit vectors in ECEF are

$$\begin{aligned} \hat{E} &= (-\sin\lambda,\ \cos\lambda,\ 0), \\ \hat{N} &= (-\sin\varphi\cos\lambda,\ -\sin\varphi\sin\lambda,\ \cos\varphi), \\ \hat{U} &= (\cos\varphi\cos\lambda,\ \cos\varphi\sin\lambda,\ \sin\varphi). \end{aligned} \tag{6}$$

A displacement (d_E, d_N, d_U) from a surveyed point (an antenna) to the physics endpoint (a detector face) is added as $\delta\vec{p} = d_E\hat{E} + d_N\hat{N} + d_U\hat{U}$. With the ellipsoid normal as $\hat{U}$ a purely vertical offset is identical to changing h by d_U. The 2014 note describes the plane tangent to a *sphere*, $\hat{U} = \vec{p}/|\vec{p}|$, which differs from the geodetic normal by the geodetic–geocentric latitude difference (0.19° at these sites); both are implemented and the difference is reported as a sensitivity.

2.5 Fermilab Site Coordinate System

Beam elements in the NuMI Technical Design Report are given in the Fermilab site system, historically "DUSAF" coordinates: x east, y north, z elevation, in US survey feet (1 ft = 1200/3937 m). Beams-doc-1148 [5] defines the modern realisation, the FSCS, as a double stereographic projection of NAD83/GRS80 geodetic coordinates: the ellipsoid is mapped conformally to a sphere and the sphere projected stereographically to the plane. This is the Oblique Stereographic projection, EPSG method 9809 [9]. The defining parameters are listed in Table 1.

Table 1: Parameters of the Fermilab Site Coordinate System [5].

origin (monument A0) latitude φ_0	N 41°50′14.312704″
origin longitude λ_0	W 88°15′41.143123″
origin ellipsoidal height h_0	186.49880 m
geodetic azimuth of the $+Y$ axis at A0, α	38°16′48.01429″
false easting, false northing	100,000 ft, 100,000 ft
scale factor at A0, 720 ft above the DUSAF datum, F_0	1.000029251309483
DUSAF → NAVD88 height correction, δH	−0.17308 m
geoid height at A0 (GEOID93), N_0	−32.78456 m
ellipsoid	GRS80 (a = 6,378,137 m, $1/f$ = 298.257222101)

With e the first eccentricity and φ_0, λ_0 the origin, the projection constants are

$$\begin{aligned}
&\rho_0 = \frac{a(1-e^2)}{(1-e^2\sin^2\varphi_0)^{3/2}}, \qquad \nu_0 = \frac{a}{\sqrt{1-e^2\sin^2\varphi_0}}, \qquad R = \sqrt{\rho_0\nu_0},\\
&n = \sqrt{1+\frac{e^2\cos^4\varphi_0}{1-e^2}}, \qquad w_1 = \left[\frac{1+\sin\varphi_0}{1-\sin\varphi_0}\left(\frac{1-e\sin\varphi_0}{1+e\sin\varphi_0}\right)^{e}\right]^{n}, \qquad \sin\chi_0' = \frac{w_1-1}{w_1+1},\\
&c = \frac{(n+\sin\varphi_0)(1-\sin\chi_0')}{(n-\sin\varphi_0)(1+\sin\chi_0')}, \qquad w_2 = c\,w_1, \qquad \chi_0 = \arcsin\frac{w_2-1}{w_2+1},
\end{aligned} \tag{7}$$

and a point (φ, λ) projects to the conformal sphere and then to the plane by

$$\begin{aligned}
&\Lambda = n(\lambda-\lambda_0)+\lambda_0, \qquad w = c\left[\frac{1+\sin\varphi}{1-\sin\varphi}\left(\frac{1-e\sin\varphi}{1+e\sin\varphi}\right)^{e}\right]^{n}, \qquad \chi = \arcsin\frac{w-1}{w+1},\\
&B = 1+\sin\chi\sin\chi_0+\cos\chi\cos\chi_0\cos(\Lambda-\lambda_0),\\
&E = \frac{2R\cos\chi\sin(\Lambda-\lambda_0)}{B}, \qquad N = \frac{2R\left[\sin\chi\cos\chi_0-\cos\chi\sin\chi_0\cos(\Lambda-\lambda_0)\right]}{B}.
\end{aligned} \tag{8}$$

The FSCS axes are rotated so that $+Y$ points along the geodetic azimuth α, rescaled to true scale at 720 ft above the DUSAF datum, and offset by the false coordinates:

$$X_{\mathrm{FSCS}} = \frac{F_0\,(E\cos\alpha - N\sin\alpha)}{1\,\mathrm{ft}} + 100{,}000\,\mathrm{ft}, \qquad Y_{\mathrm{FSCS}} = \frac{F_0\,(N\cos\alpha + E\sin\alpha)}{1\,\mathrm{ft}} + 100{,}000\,\mathrm{ft}. \tag{9}$$

Heights relate as $h = H_{\mathrm{DUSAF}} + \delta H + N_g$ with N_g taken as the A0 value across the site (the GEOID93 slope between A0 and the Main Injector centroid is 1.3 cm). The inverse of Eq. (8) is the EPSG inverse (iteration on the isometric latitude); the implementation reproduces the EPSG worked example (Amersfoort) to 2 mm and the CFMI control point of Beams-doc-1148 to 0.1 mm horizontally. A beam-sheet bearing β (degrees counter-clockwise from site East) corresponds to the geodetic azimuth $(90^\circ - \beta) + \alpha$ at A0; the NuMI final bearing 152.18332° gives 336.097°, against 336.098° from the 1999 Fermilab–Soudan geodetic survey [10, 11].

2.6 NuMI beamline system (origin MCZERO) and the beam-axis frame

The Fermilab alignment group reports NuMI positions in a Cartesian beamline system: origin MCZERO; X positive to beam right (eastward) looking downstream; Y the stationing, positive downstream; Z positive upward and orthogonal to the beam [6, 7]. The same construction serves as the beam-axis frame for off-axis angles. Given an origin O and a second point T on the line, with ECEF positions $\vec{p}_O$, $\vec{p}_T$ and the ellipsoid normal $\hat{n}_O$ at the origin,

$$\hat{u} = \frac{\vec{p}_T - \vec{p}_O}{|\vec{p}_T - \vec{p}_O|}, \qquad \hat{h} = \frac{\hat{u}\times\hat{n}_O}{|\hat{u}\times\hat{n}_O|}, \qquad \hat{v} = \hat{h}\times\hat{u}. \tag{10}$$

$\hat{u}$ is along the line, $\hat{h}$ to the right looking downstream (horizontal at the origin), $\hat{v}$ up, orthogonal to the line and in the vertical plane through $\hat{n}_O$. A point P with $\vec{r} = \vec{p}_P - \vec{p}_O$ has beamline coordinates and off-axis quantities

$$Y = \vec{r}\cdot\hat{u}, \qquad X = \vec{r}\cdot\hat{h}, \qquad Z = \vec{r}\cdot\hat{v}, \qquad \theta_{\mathrm{off}} = \arctan\frac{\sqrt{X^2+Z^2}}{Y}, \qquad \psi_{\mathrm{clock}} = \mathrm{atan2}(X, Z), \tag{11}$$

where ψ_{clock} is 0° above the axis and 90° to the right; the inverse is $\vec{p}_P = \vec{p}_O + Y\hat{u} + X\hat{h} + Z\hat{v}$. The angle between a chord $\vec{d}$ and the axis is $\angle(\vec{d}, \hat{u})$ with components $\mathrm{atan2}(\vec{d}\cdot\hat{h}, \vec{d}\cdot\hat{u})$ (right)

and atan2($\vec{d}\cdot\hat{v}, \vec{d}\cdot\hat{u}$) (up). The opening angle at a vertex V between the chords to A and B is $\angle(\vec{p}_A-\vec{p}_V, \vec{p}_B-\vec{p}_V)$, decomposed into the azimuth and elevation differences of the two chords at V; the transverse distance of A from the line VB is $|\vec{p}_A-\vec{p}_V|\sin\angle$. Angles between vectors are evaluated as atan2($|\vec{a}\times\vec{b}|, \vec{a}\cdot\vec{b}$) for numerical stability at the microradian level.

Realisation for NuMI. Two elements of the beamline system are not stated in the survey notes and were established here. (1) MCZERO lies at the upstream end of horn 1, between the target exit and the horn; it is taken as the horn-1 upstream face of the 1998 TDR beam sheet (row 0089, 1.5 m downstream of the target entrance). The as-built point is not public, so the origin is good to about 1 m along the beam, which is 0.2 ticks on an 810 km chord and nothing transverse. (2) The direction of the Y axis. Two candidate lines were tested: the 1998 design axis (target → decay-pipe end) and the as-built aiming line (MCZERO → MINOS FD midpoint). The FD survey note gives an independent NAVD88 elevation of the face centre, $H = 1{,}209.1058\,\text{ft} = 368.535\,\text{m}$, i.e. $h \simeq 336.07\,\text{m}$ with the Ash River geoid separation $N_g = -32.462\,\text{m}$ of the receiver. The aiming line reproduces it to 1.5 m; the design axis misses it by 17 m. The aiming line is therefore the surveyors' Y axis and is used for both survey points (Sec. 7).

Detector faces. A face of half-width w and half-height w' centred at (X, Y, Z) has edge points $(X \mp w, Y, Z)$ (left, right) and $(X, Y, Z \pm w')$ (top, bottom) in the same frame; the face is taken perpendicular to the beamline rather than to the detector's own axis, a 3 cm difference for the ND. The angle the face subtends at the axis origin is $\angle(\vec{r}_{\text{left}}, \vec{r}_{\text{right}})$.

3 Input positions

3.1 NuMI target, MCZERO and the design beam axis

The NuMI Technical Design Report [4], Appendix A beam sheet "horn2 ces", lists every beamline element in DUSAF coordinates (entrance of the device). The rows used are the target entrance (0087, `HALTA TAR-2`), the horn-1 entrance (0089, `IHHRN1`, the MCZERO proxy) and the decay-pipe end (0092, `DENCU`); the sheet's final bearing is 152.18332° (ccw from site East) and pitch −3.336°. These are design positions; as-built target positions in the LE and ME configurations differ at the metre level.

3.2 Surveyed upstream-face centres of the NOvA detectors

The definitive endpoints are the upstream-face centres surveyed by the Fermilab Survey, Alignment and Geodesy group [6, 7], Table 2. The FD point is 7.5 in from the intersection of the rail midline with the wall (the bookend) and vertically at the measured highest floor point (1,181.8175 ft NAVD88) plus the 23.25 in pallet and the 304.21 in half-plane, $H = 1{,}209.1058$ ft. The ND point uses the 3×3-module width and height of 158.125 in and the tunnel cross-section drawing 489389.

Table 2: Surveyed upstream-face centres in the NuMI beamline system (origin MCZERO; X right, Y stationing, Z up) [6, 7]. Frame as realised in this note.

point	X (m)	Y (m)	Z (m)	frame
NOvA ND upstream face centre (Fermilab alignment survey)	−11.565000	993.350283	−3.643551	mczero → minos-fd
NOvA FD upstream face centre (Fermilab alignment survey)	−11,037.455993	810,422.323775	−4,162.644422	mczero → minos-fd

3.3 GPS antenna fixes, MINOS far detector and other inputs

Table 3 lists the remaining inputs: the MINOS surface-building mast antenna and the three Ash River antennas from the 2014 note [1] (decoded from the TDU receivers' `$GPGGA` sentences; the Table 2 heights of the note are ellipsoidal, as the `$POLYP` altitude confirms), the MINOS FD supermodule midpoint [12] used to define the aiming line, and the 2014-note antenna-derived endpoints that are retained for comparison.

Table 3: Other input positions. FNAL antenna coordinates are quoted to 1″ (about 31 m N–S, 23 m E–W); Ash River coordinates to 10^{-6} arc-minute.

id	site	position as given	source
target	NuMI target (TDR 1998) upstream face	FSCS x, y, z = 100327.838, 97780.188, 626.743 ft	NuMI TDR 1998 App. A beam sheet (design); Beams-doc-1148 projection
mczero	MCZERO horn 1 upstream face, (TDR 1998)	FSCS x, y, z = 100324.941, 97781.717, 626.552 ft	NuMI TDR 1998 App. A beam sheet row 0089 IHHRN1
decay-end	NuMI decay pipe end (TDR 1998)	FSCS x, y, z = 98230.009, 98887.039, 488.477 ft	NuMI TDR 1998 App. A beam sheet row 0092 DENCU
fnal-mast	FNAL MINOS Surface Building mast antenna	41° 50' 23.0000" N, 88° 16' 13.0000" W, h = 228.000 m (ellipsoid)	A.Norman 2014 note Table 2 (MINOS-based TDU receivers)
nd-under-mast	ND endpoint if it were under the MINOS mast (105 + 15 m down)	41° 50' 23.0000" N, 88° 16' 13.0000" W, h = 228.000 m (ellipsoid)	2014 note sec. 3.1 and Tables 2-3 (superseded: the ND is not under the mast)
fd1	Ash River Ant-1 (FD front-face centre)	48° 22' 42.9468" N, 92° 49' 52.6158" W, h = 348.000 m (ellipsoid)	2014 note Table 2
fd2	Ash River Ant-2 fix, front face via -19.8 m (antenna-derived, superseded)	48° 22' 42.9468" N, 92° 49' 52.6158" W, h = 348.638 m (ellipsoid)	2014 note Table 2 (antenna used by the 2013 notebook)
fd3	Ash River Ant-3 (FD front-face centre)	48° 22' 42.9850" N, 92° 49' 52.6583" W, h = 350.346 m (ellipsoid)	2014 note Table 2 / $GPGGA 2013-03-07 21:03:41 UTC
minos-fd	MINOS Far Detector (supermodule midpoint, Soudan)	47° 49' 12.9612" N, 92° 14' 29.0832" W, h = −248.400 m (ellipsoid)	MINOS detectors NIM A 596 (2008) 190, sec. 6.5

3.4 Converted positions

Table 4 gives every endpoint used below in geodetic (WGS84) and ECEF coordinates after all conversions (site projection, beamline frame, antenna offsets); Table 5 repeats the principal points in decimal degrees.

4 Distances

4.1 Chords in the spherical and ellipsoidal models

Neutrinos travel the Euclidean chord $L = |\vec{p}_B - \vec{p}_A|$ through the Earth, not the surface arc. Table 6 gives, for every baseline, the chord in the spherical reference model, in the WGS84 model with the surveyed or given points, and after the antenna-to-endpoint offsets where any apply.

4.2 Correction ledger

The ledger lists each change of the endpoints in order with its effect on L, t and the tick count. For the surveyed faces only the Earth model enters (Table 7); the 2014-note geometry with its antenna

Table 4: Endpoints in WGS84 geodetic and ECEF coordinates. `nd/fd` are the surveyed faces, `-left/-right` their edges; `nd-1015`, `nd-under-mast` and `fd2` are superseded geometries (Appendix A).

id	latitude	longitude	h (m)	X (m)	Y (m)	Z (m)
target	41° 49' 55.0917" N	88° 15' 55.9086" W	158.074	144,061.137	−4,757,403.941	4,231,821.378
mczero	41° 49' 55.1213" N	88° 15' 55.9261" W	158.016	144,060.713	−4,757,403.302	4,231,822.020
decay-end	41° 50' 16.5133" N	88° 16' 08.6061" W	115.930	143,754.011	−4,756,940.784	4,232,285.708
nd	41° 50' 24.3467" N	88° 16' 13.7981" W	96.572	143,628.970	−4,756,768.845	4,232,452.856
nd-left	41° 50' 24.3203" N	88° 16' 13.8777" W	96.572	143,627.151	−4,756,769.443	4,232,452.250
nd-right	41° 50' 24.3731" N	88° 16' 13.7185" W	96.572	143,630.788	−4,756,768.247	4,232,453.463
fd	48° 22' 41.5413" N	92° 49' 50.7737" W	337.611	−209,624.380	−4,239,417.585	4,745,165.129
fd-left	48° 22' 41.4272" N	92° 49' 51.1078" W	337.623	−209,631.378	−4,239,419.886	4,745,162.796
fd-right	48° 22' 41.6555" N	92° 49' 50.4396" W	337.599	−209,617.382	−4,239,415.284	4,745,167.462
minos-fd	47° 49' 12.9612" N	92° 14' 29.0832" W	−248.400	−167,796.989	−4,287,098.683	4,703,296.906
nd-1015	41° 50' 24.9725" N	88° 16' 14.1694" W	104.124	143,620.189	−4,756,761.853	4,232,472.279
nd-under-mast	41° 50' 23.0000" N	88° 16' 13.0000" W	108.000	143,648.469	−4,756,804.502	4,232,429.524
fd2	48° 22' 42.9468" N	92° 49' 52.6158" W	328.838	−209,660.350	−4,239,377.478	4,745,187.407

Table 5: Principal points in decimal degrees (WGS84).

id	latitude (°)	longitude (°, east +)	h (m)
target	41.831969923	−88.265530166	158.074
mczero	41.831978142	−88.265535036	158.016
nd	41.840096298	−88.270499477	96.572
fd	48.378205927	−92.830770475	337.611
minos-fd	47.820267000	−92.241412000	−248.400

Table 6: Chord length by Earth model. L_{sph}: sphere of radius a; L_{WGS}: ellipsoid, points as given; L_{final}: after antenna offsets (none for surveyed faces).

baseline	L_{sph} (m)	L_{WGS} (m)	Δ (m)	L_{final} (m)	Δ (m)
ND upstream face → FD upstream face (both surveyed)	810,337.734	809,514.750	−822.984	809,514.750	0.000
NuMI target → FD upstream face (surveyed)	811,333.740	810,509.171	−824.569	810,509.171	0.000
NuMI target → ND upstream face (surveyed)	996.010	994.424	−1.585	994.424	0.000
MCZERO → FD upstream face (surveyed; stationing 810,422.324 m)	811,332.738	810,508.171	−824.567	810,508.171	0.000
MCZERO → ND upstream face (surveyed; stationing 993.350 m)	995.008	993.424	−1.584	993.424	0.000
ND at the quoted 1015 m → FD (superseded ND placement)	810,317.097	809,494.148	−822.949	809,494.148	0.000
ND under the mast → FD from the Ant-2 fix (superseded 2014-note geometry)	810,448.173	809,625.125	−823.048	809,616.263	−8.862
NuMI target → MINOS FD (validation of the target position)	736,157.346	735,344.614	−812.732	735,344.614	0.000
NuMI target → decay-pipe end (TDR design axis)	725.363	724.184	−1.179	724.184	0.000

offsets is given for comparison in Table 24 of Appendix A.

Table 7: Correction ledger, ND face → FD face.

step	applied	L (m)	ΔL (m)	Δt (ns)	Δticks
reference model	spherical Earth, R = 6378137.000 m; heights as given; surveyed points, no offsets	810,337.734	0.000	0.00	0.000
Earth model	WGS84 ellipsoid; heights as given; surveyed points, no offsets	809,514.750	−822.984	−2,745.18	−175.691

4.3 Chord geometry

The geocentric central angle $\gamma = \angle(\vec{p}_A, \vec{p}_B)$, the surface arc at the mean geocentric radius $\bar{R}\gamma$, the sagitta $\bar{R}(1 - \cos\gamma/2)$, and the azimuth and elevation of the chord in the local frames of Eq. (6) at each end are given in Table 8. A great-circle distance would be wrong by more than 100 ticks.

Table 8: Chord geometry. Elevation is negative when the chord dips below the local horizontal at the source; at the target end a positive value means the chord arrives rising.

chord	γ (°)	arc (m)	arc−chord (m)	sagitta (m)	azimuth (°)	dip at source (°)	elev. at target (°)	deepest h (m)
ND → FD	7.28884	810,057.9	543.16	12,877.0	335.30891	−3.62569	3.65835	−12,651
target → FD	7.29777	811,054.2	545.00	12,908.6	335.31237	−3.63452	3.65845	−12,652
target → ND	0.00893	992.3	−2.09	0.0	335.42658	−3.55027	−3.54134	97
target → MINOS FD	6.62038	735,749.1	404.52	10,623.8	336.09421	−3.34056	3.27617	−10,664

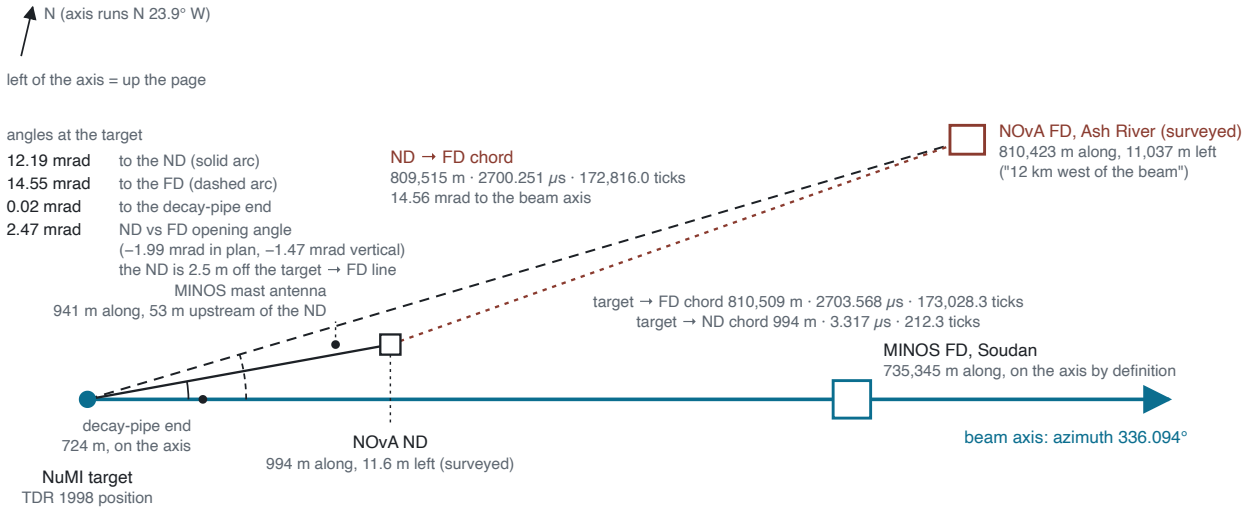


Figure 1: Plan view, not to scale: the beam axis from the NuMI target through the MINOS far detector, the surveyed ND and FD faces to its left, the MINOS mast antenna, and the chords used in the calculation.

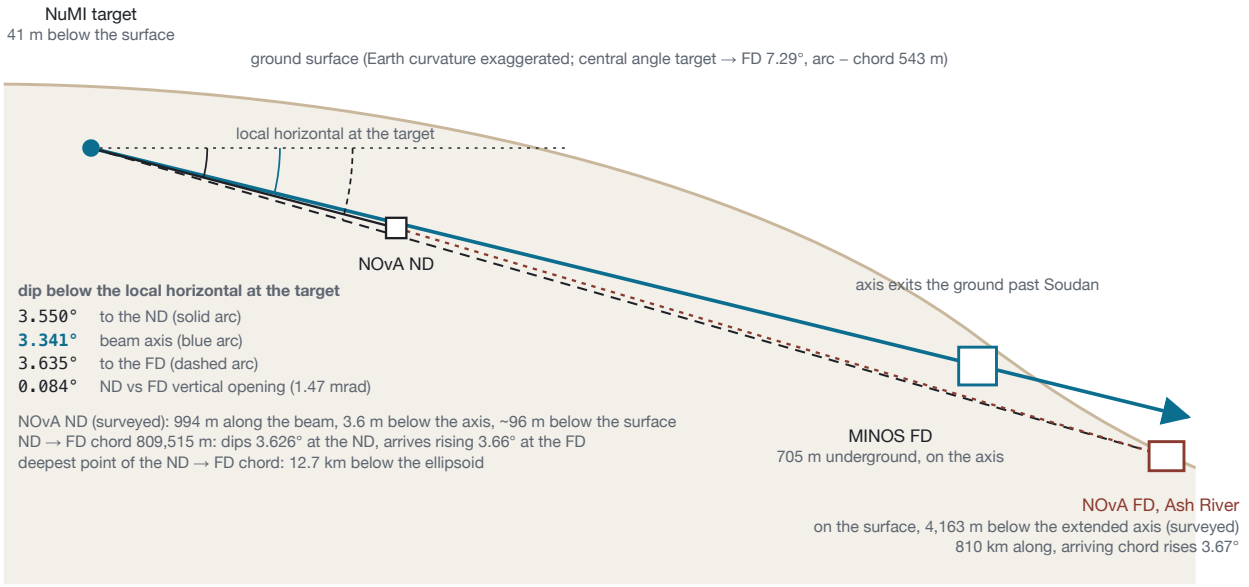


Figure 2: Elevation view, not to scale: dips of the chords below the target's local horizontal, the curved ground surface, and the positions of the detectors relative to the extended beam axis.

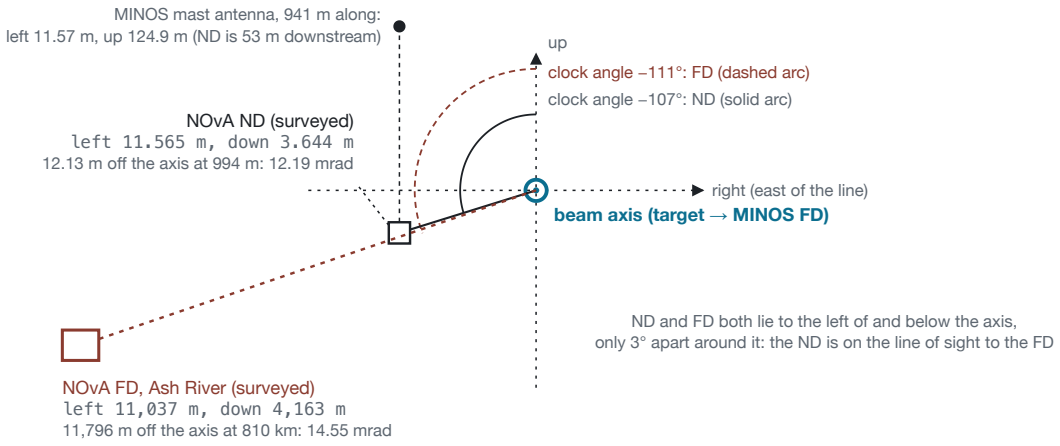


Figure 3: Transverse view looking downstream along the beam axis: the surveyed ND and FD faces lie to the left of and below the axis at almost the same clock angle.

5 Time of flight and timing offsets

5.1 Flight time and clock ticks

With $c = 299{,}792{,}458\,\mathrm{m/s}$ exactly and neutrinos taken to travel at c (the mass-induced delay is evaluated in Sec. 5.4),

$$t = \frac{L}{c}, \qquad n = f_{\mathrm{clk}}\, t, \qquad f_{\mathrm{clk}} = 64\,\mathrm{MHz}, \tag{12}$$

so one tick is 15.625 ns or 4.684 m of light path. The legacy Global Trigger unit of 500 ns is also given. Table 9 lists the results for the surveyed geometry; the integer tick count uses round-to-nearest and the residual is the difference between the integer and the exact value.

Table 9: Flight times for the surveyed geometry. "total" includes the Main Injector shift of Sec. 5.2.

chord	L (m)	t (µs)	ticks	integer	residual (ns)	500 ns units	total ticks
ND upstream face → FD upstream face (both surveyed)	809,514.750	2,700.250552	172,816.035	172,816	−0.55	5,400.5011	272,736.417
NuMI target → FD upstream face (surveyed)	810,509.171	2,703.567584	173,028.325	173,028	−5.08	5,407.1352	272,948.707
NuMI target → ND upstream face (surveyed)	994.424	3.317043	212.291	212	−4.54	6.6341	100,132.672
MCZERO → FD upstream face (surveyed; stationing 810,422.324 m)	810,508.171	2,703.564248	173,028.112	173,028	−1.75	5,407.1285	272,948.494
MCZERO → ND upstream face (surveyed; stationing 993.350 m)	993.424	3.313707	212.077	212	−1.21	6.6274	100,132.459

5.2 Main Injector timing shift

After the 2012 shutdown the beam moved relative to the Main Injector Beam Sync event MIBS \$74 by 141 MI revolutions [3]. With $h_{\mathrm{MI}} = 588$ RF buckets per turn at the flat-top RF frequency f_{RF},

$$T_{\mathrm{rev}} = \frac{h_{\mathrm{MI}}}{f_{\mathrm{RF}}}, \qquad \Delta t_{\mathrm{MI}} = 141\, T_{\mathrm{rev}}, \qquad n_{\mathrm{MI}} = f_{\mathrm{clk}}\, \Delta t_{\mathrm{MI}}, \tag{13}$$

giving Table 10. One revolution is 708.655 ticks: the 53.1034 MHz RF and the 64 MHz clock are not commensurate, and a 1 kHz error in f_{RF} shifts n_{MI} by 1.882 ticks.

Table 10: Main Injector 141-turn shift.

turns	buckets/turn	f_{RF} (MHz)	T_{rev} (µs)	ticks/turn	Δt_{MI} (µs)	n_{MI} (ticks)
141	588	53.1034	11.072737	708.655	1,561.255965	99,920.382

5.3 Combined offset and trigger window

The total offset of the FD trigger relative to MIBS \$74 is $n_{\mathrm{tot}} = n_{\mathrm{ToF}} + n_{\mathrm{MI}}$ (Table 11); the Sagnac term is reported but not added. The window of Eq. (1) in ticks is given in Table 12; a 50 µs window spans 3,200 ticks and overlaps one or two 50 µs microslices, so a recorded beam event is up to 100 µs wide.

Table 11: Combined offset, ND face → FD face plus the MI shift.

n_{ToF}	n_{MI}	n_{tot}	integer	residual (ns)
172,816.035	99,920.382	272,736.417	272,736	−6.52

Table 12: Trigger window $[t - 20\,\mu\mathrm{s},\ t + 30\,\mu\mathrm{s})$ in ticks after t_0, relative to the flight time alone and to the flight time plus the MI shift.

ToF low	ToF high	total low	total high	ticks/microslice	microslices overlapped
171,536.04	174,736.04	271,456.42	274,656.42	3,200	1–2

5.4 Effects outside the geometric flight time

In the rotating ECEF frame a signal from A to B acquires the Sagnac correction $\Delta t_{\mathrm{S}} = \omega_{\oplus}\,(x_A y_B - y_A x_B)/c^2$ with $\omega_{\oplus} = 7.2921150 \times 10^{-5}\,\mathrm{rad/s}$ [13]; it is negative here because the beam travels westward and the FD rotates toward it. A neutrino of mass m and energy E is delayed by $\delta t = t\,m^2/(2E^2)$. Antenna cable delays ($\ell/(v_f c)$) are receiver calibrations and not part of the flight time. Table 13 lists them; the MINOS neutrino-speed measurement [14] treats the same terms.

Table 13: Non-geometric terms for the ND → FD chord.

effect	ns	ticks	status
Sagnac (Earth rotation, ECEF frame)	−1.303	−0.0834	reported, not applied
neutrino mass delay ($m = 0.1\,\mathrm{eV}$, $E = 2\,\mathrm{GeV}$)	3.38×10^{-15}	2.16×10^{-16}	negligible
antenna cable delay, NOvA ND upstream face centre (Fermilab alignment survey)	119.61	7.655	receiver calibration, not ToF
antenna cable delay, NOvA FD upstream face centre (Fermilab alignment survey)	119.61	7.655	receiver calibration, not ToF

6 Angles relative to the beamline

6.1 Beam axis and off-axis positions

The beam axis of this note is the line from the target through the MINOS FD midpoint (Table 14); the TDR design axis agrees with it in direction to 0.02 mrad. Table 15 gives every site's off-axis angle at the target, distance along the axis and transverse position from Eq. (11).

6.2 Opening angles

Table 16 gives the opening angles at the target between pairs of chords, their horizontal and vertical components, the transverse distance of the first endpoint from the second chord, and each chord's angle to the beam axis. With both surveyed faces the ND lies only 2.46 m off the target → FD line: the Near Detector sits on the line of sight to the Far Detector.

Table 14: The beam axis.

axis	length (m)	azimuth at origin (°)	elevation at origin (°)	unit vector (ECEF)
NuMI beam axis (target → MINOS FD)	735,344.614	336.09421	−3.34056	(-0.424097927, 0.639571229, 0.641162687)

Table 15: Sites relative to the beam axis: off-axis angle at the target, distance along the axis, transverse distance with right/up components, and clock angle (0° above, 90° right).

id	off-axis (mrad)	along (m)	transverse (m)	right (m)	up (m)	clock (°)
target	0.0000	0.000	0.000	0.000	0.000	–
mczero	0.1459	1.000	0.000	0.000	0.000	64.88
decay-end	0.0197	724.184	0.014	0.000	0.014	−1.89
nd	12.1935	994.350	12.125	−11.565	−3.643	−107.49
fd	14.5547	810,423.324	11,796.315	−11,037.456	−4,162.644	−110.66
minos-fd	0.0000	735,344.614	0.000	0.000	0.000	–
nd-1015	12.4658	1,015.000	12.653	−11.570	5.123	−66.12
nd-under-mast	13.3426	948.316	12.654	−11.570	5.123	−66.12
fd2	14.5753	810,478.514	11,813.783	−11,051.360	−4,175.273	−110.70
fnal-mast	132.4921	941.341	125.455	−11.570	124.921	−5.29

Table 16: Opening angles at the NuMI target.

angle	total (mrad)	horizontal (mrad)	vertical (mrad)	a transverse to Vb (m)	a vs axis (mrad)	b vs axis (mrad)
Target: ND vs FD (both surveyed)	2.4740	−1.9895	−1.4705	2.460	12.1935	14.5547
Target: ND (surveyed) vs MINOS FD	12.1935	11.6313	3.6600	12.125	12.1935	0.0000
Target: ND at the quoted 1015 m vs surveyed ND	8.7139	−0.2374	−8.7107	8.845	12.4658	12.1935
Target: ND under the mast (2014 note) vs surveyed ND	9.0840	0.5638	−9.0665	8.615	13.3426	12.1935
Target: FD from the Ant-2 fix vs surveyed FD	0.0223	0.0162	0.0152	18.037	14.5753	14.5547

6.3 Near Detector face edges

The ND face is 158.125 in square (half-width 2.0081875 m). Table 17 gives the positions of its centre and edges and their angles to the beam axis; the face subtends 4.0386 mrad (left–right) and 4.0389 mrad (top–bottom) at the target. Table 18 gives the opening angle between each face point and the target → FD chord. Figure 4 shows the face with the iso-angle contours. The FD face (half-size 304.21 in) is treated identically; its edges are 0.019 mrad apart as seen from the target.

Table 17: ND face points: geodetic position, angle to the beam axis and transverse offsets from the axis.

point	latitude	longitude	h (m)	angle to axis (mrad)	right (m)	up (m)	clock (°)
left	41° 50' 24.3203" N	88° 16' 13.8777" W	96.572	14.1325	−13.573	−3.643	−105.03
center	41° 50' 24.3467" N	88° 16' 13.7981" W	96.572	12.1935	−11.565	−3.643	−107.49
right	41° 50' 24.3731" N	88° 16' 13.7185" W	96.572	10.2854	−9.557	−3.643	−110.87
top	41° 50' 24.3501" N	88° 16' 13.8002" W	98.577	11.7457	−11.565	−1.635	−98.05
bottom	41° 50' 24.3432" N	88° 16' 13.7961" W	94.568	12.9444	−11.565	−5.652	−116.04

Table 18: Opening angle at the target between each ND face point and the chord to the FD face.

point	opening angle (mrad)	horizontal (mrad)	vertical (mrad)	transverse to the FD chord (m)	chord vs axis (mrad)
left	1.4724	0.0296	−1.4721	1.464	14.1325
center	2.4740	−1.9895	−1.4705	2.460	12.1935
right	4.2695	−4.0088	−1.4691	4.246	10.2854
top	4.0179	−1.9910	−3.4899	3.995	11.7457
bottom	2.0624	−1.9880	0.5489	2.051	12.9444

7 Validation

The target position and the realisation of the beamline frame rest on design documents; Table 19 checks them against independently published or surveyed quantities. The target → MINOS FD chord agrees with the published 735,338.0 m (from the horn-1 upstream end) to 7 m, the beam azimuth with the 1999 geodetic survey to 13″, the FD ellipsoidal height with the NAVD88 survey elevation to 1.5 m, and the MCZERO → FD chord equals $\sqrt{X^2+Y^2+Z^2}$ of the survey coordinates identically. The surveyed ND confirms the 2014 antenna's 11.57 m left offset to 5 mm. In addition the 2013 notebook is reproduced to 10^{-9} (Appendix A).

8 Sensitivities and error budget

Numerical partial derivatives of the ND → FD chord with respect to each input are given in Table 20; the gradient of L with respect to an endpoint position is the unit chord vector, so an endpoint error of σ metres along the chord is worth σ metres of L (3.34 ns/m). The error budget (Table 21) combines the declared 1σ uncertainties in quadrature and lists conventions and neglected terms separately. With both faces surveyed, the budget is dominated by the RF-frequency term of the Main Injector shift; combined items: root-sum-square 1.906 ticks (29.8 ns), linear sum 2.309 ticks (36.1 ns) .

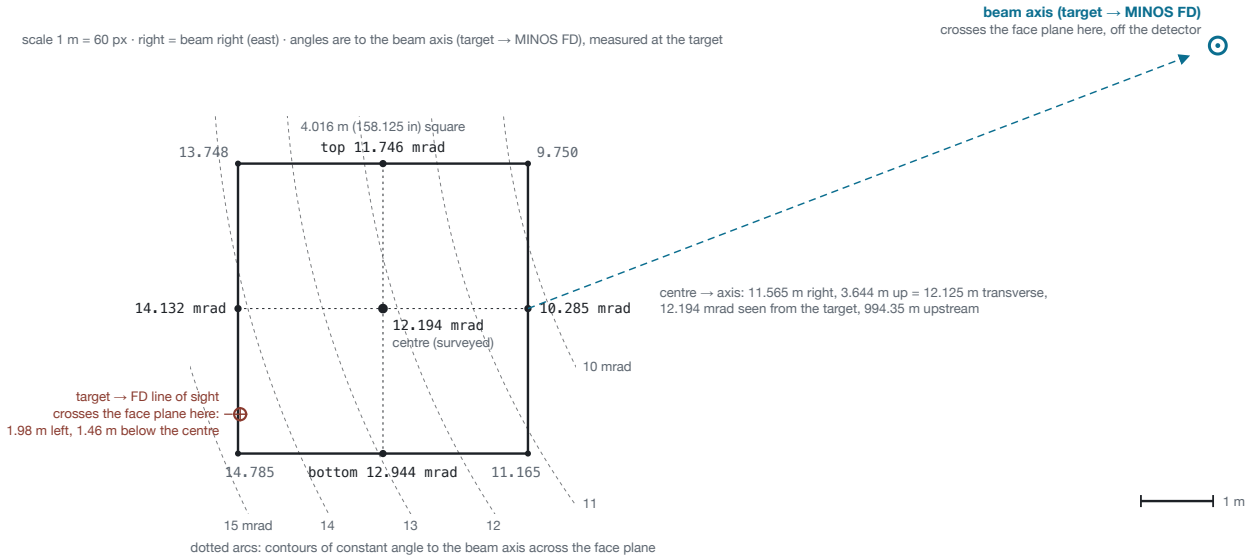


Figure 4: The ND front face looking downstream, to scale: the angle to the beam axis at the centre, edges and corners, contours of constant angle across the face plane, the point where the beam axis crosses the face plane (off the detector, to the right and above), and the point where the target → FD line of sight crosses the face plane.

Table 19: Validation against published and surveyed quantities.

quantity	this note	reference	difference
target → MINOS FD midpoint	735,344.6 m	735,338.0 m from horn 1 [12]	6.6 m
beam azimuth at the target	336.09421°	336.09788° [10]	−0.00367°
beam dip at the target	3.34056°	3.34349° [15]	−0.00293°
decay-pipe length (TDR stations)	675.0 m	675 m [15]	–
FD ellipsoidal height from the survey frame	337.611 m	$H + N = 368.535 - 32.462 =$ 336.073 m	1.54 m
MCZERO → FD chord vs $\lvert(X, Y, Z)\rvert$	810,508.1714 m	810,508.1714 m	0.000000 m
off-axis angle of the FD face	14.555 mrad	14.6 mrad [16]	–

Table 20: Sensitivities of the ND → FD chord.

variation	ΔL (m)	Δt (ns)	Δticks
NOvA ND upstream face centre (Fermilab alignment survey): height +1 m	0.0632	0.211	0.0135
NOvA ND upstream face centre (Fermilab alignment survey): latitude +1"	−27.9763	−93.319	−5.9724
NOvA ND upstream face centre (Fermilab alignment survey): longitude +1"	9.6187	32.085	2.0534
NOvA ND upstream face centre (Fermilab alignment survey): endpoint +1 m East	0.4169	1.391	0.0890
NOvA ND upstream face centre (Fermilab alignment survey): endpoint +1 m North	−0.9068	−3.025	−0.1936
NOvA FD upstream face centre (Fermilab alignment survey): height +1 m	0.0638	0.213	0.0136
NOvA FD upstream face centre (Fermilab alignment survey): latitude +1"	27.2369	90.852	5.8146
NOvA FD upstream face centre (Fermilab alignment survey): longitude +1"	−9.6182	−32.083	−2.0533
NOvA FD upstream face centre (Fermilab alignment survey): endpoint +1 m East	−0.4674	−1.559	−0.0998
NOvA FD upstream face centre (Fermilab alignment survey): endpoint +1 m North	0.8817	2.941	0.1882
spherical Earth (R = 6378137 m) at the final endpoints	822.9838	2,745.178	175.6914
GRS80 ellipsoid instead of WGS84	0.0000	0.000	0.0000

Table 21: Error budget, ND face → FD face, in 64 MHz ticks.

source	ns	ticks	type	combined
NOvA ND upstream face centre (Fermilab alignment survey): endpoint offset ±1 m, worst case along the chord	3.34	0.213	systematic	yes
NOvA FD upstream face centre (Fermilab alignment survey): endpoint offset ±1 m, worst case along the chord	3.34	0.213	systematic	yes
Main Injector 141-turn shift (2012 shutdown): RF frequency ±1000 Hz	29.40	1.882	systematic	yes
integer rounding of the final tick count	7.81	0.500	convention	no
Sagnac term (not applied)	1.30	0.083	neglected	no

9 Summary

Table 22 collects all baselines. For the surveyed ND and FD faces the flight path is 809,514.750 m, the flight time 2,700.250552 µs = 172,816.035 ticks, and the total offset with the 141-turn Main Injector shift 272,736.417 ticks, i.e. 272,736 ticks when rounded (residual −6.5 ns). From the target the FD face is 810,509.171 m (173,028.325 ticks) and the ND face 994.424 m (212.291 ticks). The ND and FD are 12.19 and 14.55 mrad off the Soudan axis, on the same side (clock −107° and −111°), and the ND face spans 4.04 mrad at the target. Open items are the as-built MCZERO and target positions (about 1 m along the beam) and the geoid model behind the 1.5 m residual on the FD height.

Table 22: All baselines of the configuration. "total" adds the MI shift; the last column is the opening angle between the chord and the beam axis.

baseline	L (m)	t (µs)	ticks	total ticks	integer	chord vs axis (mrad)
ND upstream face → FD upstream face (both surveyed)	809,514.750	2,700.250552	172,816.035	272,736.417	272,736	14.5576
NuMI target → FD upstream face (surveyed)	810,509.171	2,703.567584	173,028.325	272,948.707	272,949	14.5547
NuMI target → ND upstream face (surveyed)	994.424	3.317043	212.291	100,132.672	100,133	12.1935
MCZERO → FD upstream face (surveyed; stationing 810,422.324 m)	810,508.171	2,703.564248	173,028.112	272,948.494	272,948	14.5547
MCZERO → ND upstream face (surveyed; stationing 993.350 m)	993.424	3.313707	212.077	100,132.459	100,132	12.2059
ND at the quoted 1015 m → FD (superseded ND placement)	809,494.148	2,700.181830	172,811.637	272,732.019	272,732	14.5618
ND under the mast → FD from the Ant-2 fix (superseded 2014-note geometry)	809,616.263	2,700.589164	172,837.706	272,758.088	272,758	14.5812
NuMI target → MINOS FD (validation of the target position)	735,344.614	2,452.845607	156,982.119	256,902.501	256,903	0.0000
NuMI target → decay-pipe end (TDR design axis)	724.184	2.415619	154.600	100,074.981	100,075	0.0197

A History and reproduction of the earlier calculations

The 2013 notebook [3] used the MINOS mast antenna (41°50′23″ N, 88°16′13″ W, 228 m) lowered by 300 m and Ash River antenna 2 (348.638 m); the 2014 note [1] specified 105 + 15 m and 12 + 7.8 m vertical offsets instead and left the horizontal antenna-to-detector vectors to be determined. Table 23 reproduces the notebook's printed outputs with the present code, and Table 24 shows the 2014-note ledger. Table 25 traces the ND → FD tick count through the successive geometries: the 2014 antenna-derived endpoints, the Near Detector placed at a quoted 1,015 m from the target with the antenna's transverse position, and the surveyed faces. The surveyed FD face is 58 m horizontally and 8.8 m vertically from the antenna-derived point; the surveyed ND face is 993.35 m from MCZERO, 3.6 m below the axis, not 5.1 m above it.

B Software and reproducibility

All numbers were produced by `gps-tof` [2] (Python, standard library plus PyYAML) from the configuration `config/nova.yaml`, which records every input with its source. The package implements Eqs. (4)–(11), the correction ledger, the sensitivities and the error budget, and is covered by a test suite that pins the reproduction of the 2013 notebook, the EPSG worked example, the Beams-doc-1148 control points, the survey coordinates and the headline results. To regenerate the report and this note's tables:

Table 23: Reproduction of the 2013 notebook [3]; the last row is the 2014-note offset geometry.

quantity	recomputed	notebook	relative difference
2013 notebook, spherical chord (m)	810,429.1726	810,429.1726	7.2×10^{-16}
2013 notebook, WGS84 chord (m)	809,606.1622	809,606.1622	5.8×10^{-16}
2013 notebook, ToF (µs)	2,700.555470	2,700.555470	6.7×10^{-16}
2013 notebook, ToF (ticks)	172,835.5501	172,835.5501	5.1×10^{-16}
2013 notebook, MI shift (ticks)	99,920.3817	99,920.3817	1.5×10^{-16}
2013 notebook, total (ticks)	272,755.9318	272,755.9318	2.1×10^{-16}
2014 note offsets, WGS84 chord (m)	809,616.2634	–	–

Table 24: Correction ledger of the superseded 2014-note geometry (antenna fixes with vertical offsets).

step	applied	L (m)	ΔL (m)	Δt (ns)	Δticks
reference model	spherical Earth, R = 6378137.000 m; heights as given; surveyed points, no offsets	810,448.173	0.000	0.00	0.000
Earth model	WGS84 ellipsoid; heights as given; surveyed points, no offsets	809,625.125	−823.048	−2,745.39	−175.705
vertical offset	ND endpoint if it were under the MINOS mast (105 + 15 m down): endpoint -120.000 m along the local geodetic vertical (105 m cavern depth + 15 m mast)	809,617.527	−7.598	−25.35	−1.622
vertical offset	Ash River Ant-2 fix, front face via -19.8 m (antenna-derived, superseded): endpoint -19.800 m along the local geodetic vertical (12 m antenna → detector top + 7.8 m to mid-height)	809,616.263	−1.263	−4.21	−0.270

Table 25: ND → FD tick count through the successive geometries.

geometry	ticks	vs surveyed
2013 notebook: antenna −300 m, no FD offset	172,835.550	19.515
2014 note: antenna −120 m, FD antenna −19.8 m	172,837.706	21.671
ND at the quoted 1015 m, antenna transverse	172,811.637	−4.398
surveyed faces (this note)	172,816.035	0.000

```
gps-tof -c config/nova.yaml -f all -o reports/nova
python3 scripts/make_note_tables.py
cd docs/note && latexmk -pdf nova-tof-note
```

Local copies of every cited document are kept in `docs/external/` and `docs/reference/` of the repository, with SHA-256 checksums and an index of provenance.

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